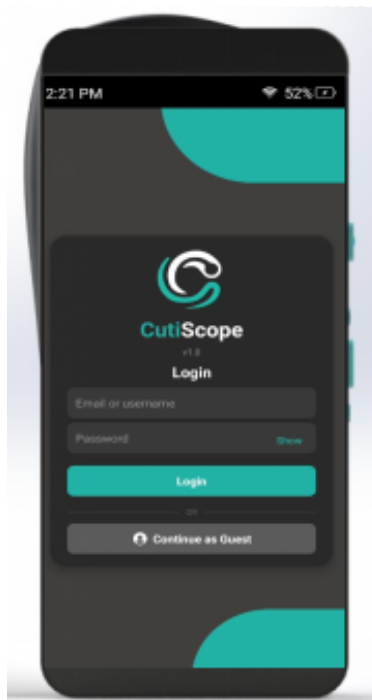


CUTISCOPE USER MANUAL

Model No: DRM100



Read all instructions and safety warnings before use. Ensure the device is fully charged and disinfected before its first clinical application.

Table of Contents

1. Introduction

The Revive Meditech **Cutiscope** is a handheld contact-based hybrid dermatoscope designed to provide enhanced visualization and analysis of skin structures using both polarized and non-polarized illumination.

The system includes:

- Dermatoscope device
- Integrated optical system (polarized & non-polarized modes)
- Image capture software
- Rechargeable battery with USB-C charging
- Simple push-button controls
- CutiScope Cloud

With up to **10x optical and 3x digital magnification**, the Cutiscope allows precise observation of skin features.

1.1. Intended Use

The device is intended for non-invasive examination of skin lesions and image capturing to assist healthcare professionals in dermatological evaluation.

1.2. User Profile

This device is intended for use by:

- Dermatologists
- Physicians
- Trained healthcare professionals

1.3. **Warnings:**

- Do not use the device for diagnosis without proper clinical training or supervision
- Do not use the device with wet hands.
- Do not use the device on open wounds, infected skin, or highly sensitive areas unless advised by a qualified healthcare professional.
- Avoid direct exposure of the device light to eyes, as it may cause discomfort or injury.

- Do not attempt to disassemble or modify the device; this may result in malfunction or electrical hazards.
- Use only the provided or recommended charging accessories to prevent damage or risk of fire.
- Do not operate the device in wet or highly humid environments to avoid electrical hazards.
- Ensure the device is properly cleaned before and after each use to prevent cross-contamination.
- Do not operate the device if the device overheats, emits unusual odor, or shows signs of damage.
- Keep the device out of reach of children and untrained individuals.

1.4. Safety Precautions:

Category	Precaution
General Use	• Operate the device only as per the user manual instructions.
Handling	• Handle the device carefully to avoid drops or physical damage.
Hygiene	• Clean and disinfect the contact surface before and after each use.
Charging	• Use only approved USB-C chargers and avoid overcharging. • Do not operate the device while it is charging.
Environment	• Use the device in a clean, dry, and well-lit environment.
Storage	• Store the device in a protective case when not in use.
Temperature	• Avoid exposure to extreme heat or cold conditions.
Battery Safety	• Do not puncture, crush, or expose the battery to high temperatures.
Light Safety	• Avoid prolonged exposure to light in the eyes or sensitive skin areas. • Do not use it for eye examinations. • Ensure patients keep their eyes closed during facial use.

Infection Control	Use proper sterilization protocols to prevent cross-patient contamination
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2. Technical Specifications

Item	Specification
Net Weight/ Dimension	g/ 175x84x45 mm
Package Weight/ Dimensions	g/ 200x105x67 mm
Operating and Storage Conditions	Room Temperature and Pressure
Light Source	LEDs
LED Type	SMD LED Beads
Polarization	15 LEDs (10 polarizing and 5 non-polarizing)
Aperture	45 mm
Digital Magnification	3x
Optical Magnification	10x
Function buttons	Push the button to switch between polarization modes of the LED Power ON/OFF Push button
Examination Method	Contact only
Battery	Rechargeable Lithium-Ion Battery

Charging Time	Min. 3 hours
Continuous duty time	Approx. 2.5-3 hours
Charging Terminal	USB Type C
Wifi Support	Yes

3. Device Components

Ensure the following items are included:

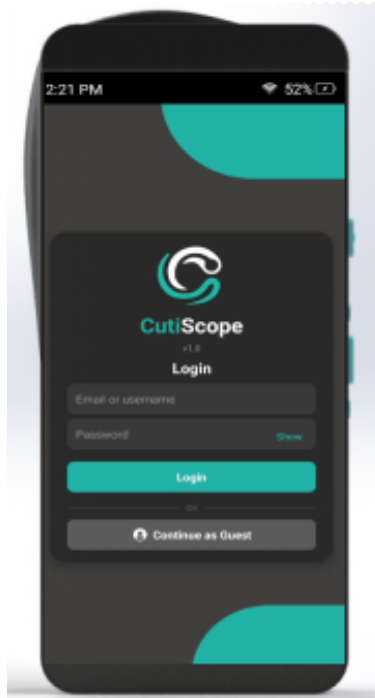
- Dermatoscope Unit
- USB Type-C charging cable
- Power adapter
- User manual

If any component is missing, contact the manufacturer.

3.1. Device Structure:



4. Device Overview



4.1. Device Operating Instructions

The device should be used according to proper operating guidelines to ensure safe and efficient performance. Follow the steps below:

- Turn on the device using the power button
- Ensure the device is fully charged before use
- Open the required application or system interface
- Perform the required operations through the screen or controls
- Turn off the device after completing usage to save battery

4.2 Device Charging and Disinfection

Proper charging and cleaning of the device is necessary for long-term durability and hygiene.

- Charge the device using the recommended charger only
- Avoid overcharging to prevent battery damage
- Keep the charging port clean and dust-free
- Disinfect the device regularly using a soft cloth and approved disinfectant
- Avoid excessive liquid contact with the device

Caution and Safety Instructions

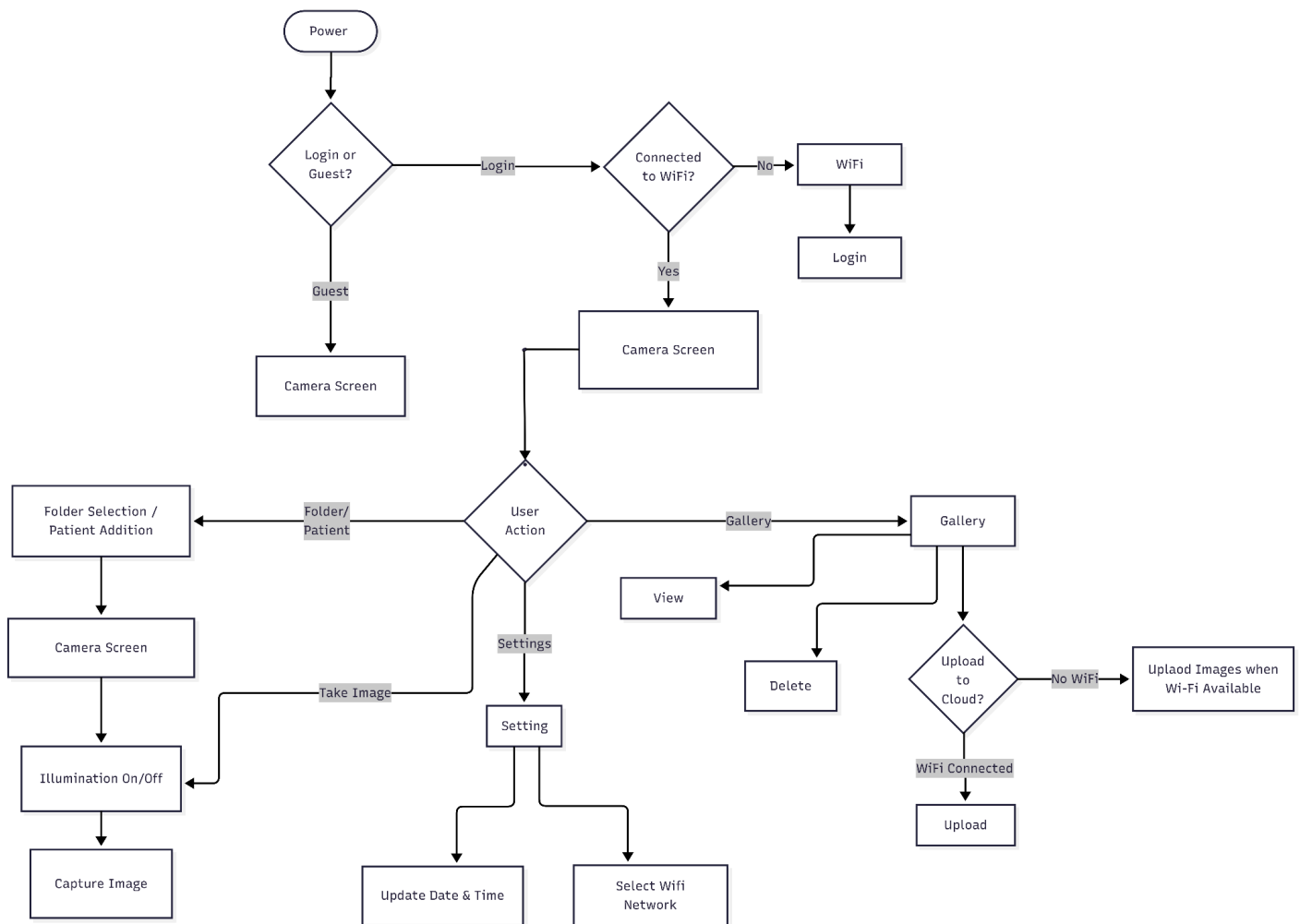
To ensure safe usage of the device, the following precautions must be followed:

- Do not use the device with wet hands
- Avoid exposure to extreme heat, fire, or direct sunlight
- Do not open or repair the device without professional assistance
- Use only original or approved accessories
- Keep the device away from children if not supervised
- Handle the device carefully to avoid physical damage
- Do not operate the device while it is charging

5. System Overview:

The Cutiscope Device operates on **software** designed to ensure reliability and focus during examinations connected with a cloud storage **web application**.

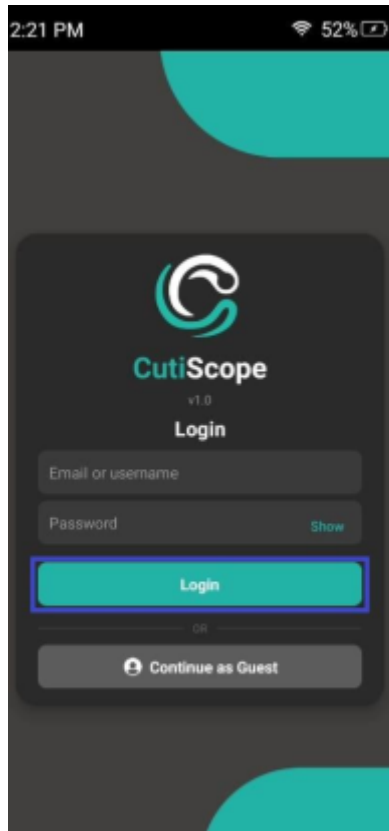
- **Unified Operational Environment:** The system is dedicated exclusively to CutiScope Dermatoscope image capture software.
- **Auto-Standby & Intelligent Wake:** To preserve power and clinician privacy, the CutiScope Dermatoscope enters a standby state after 2 minutes of inactivity.



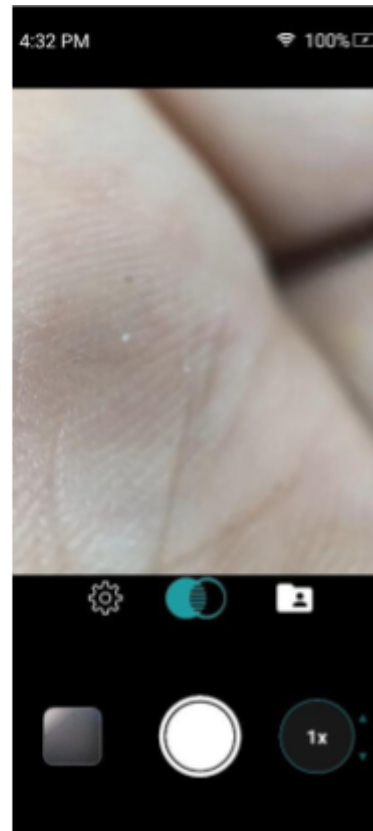
5.1. Authentication & Security

5.1.1. Clinical Sign-In:

Authorized medical professionals must authenticate their credentials. This ensures every diagnostic capture is linked to a specific clinician for accurate record-keeping and auditing.



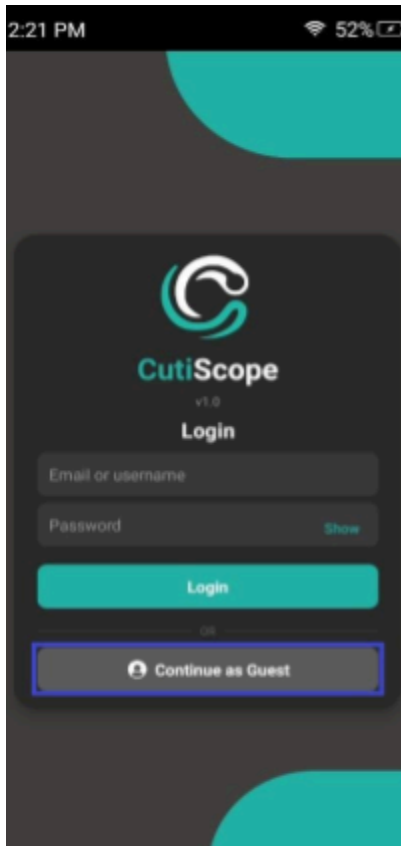
Login as Clinician



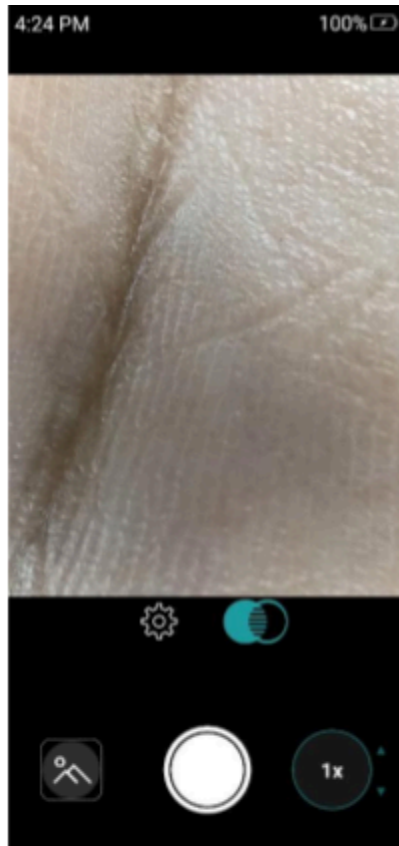
Camera Screen

5.1.2. Guest Session Protocol:

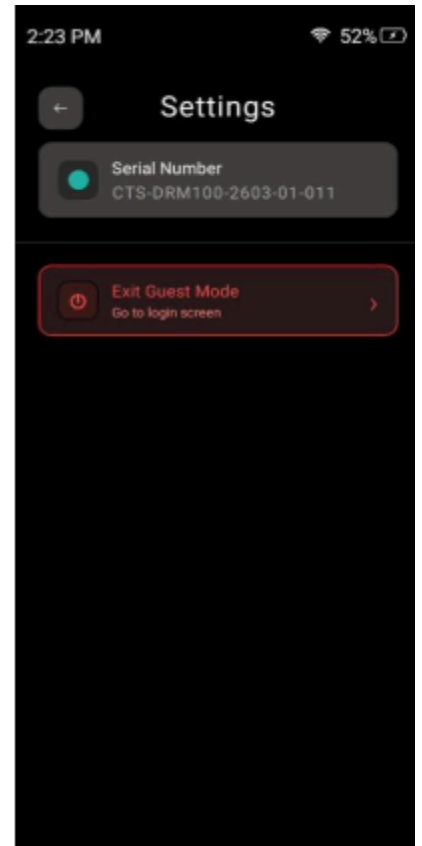
A temporary Guest session is available for immediate use without a permanent account. The images that are captured during the guest mode will be saved temporarily and will be removed upon exit.



Continue as Guest



Camera Screen



Settings Screen

⚠ CAUTION:

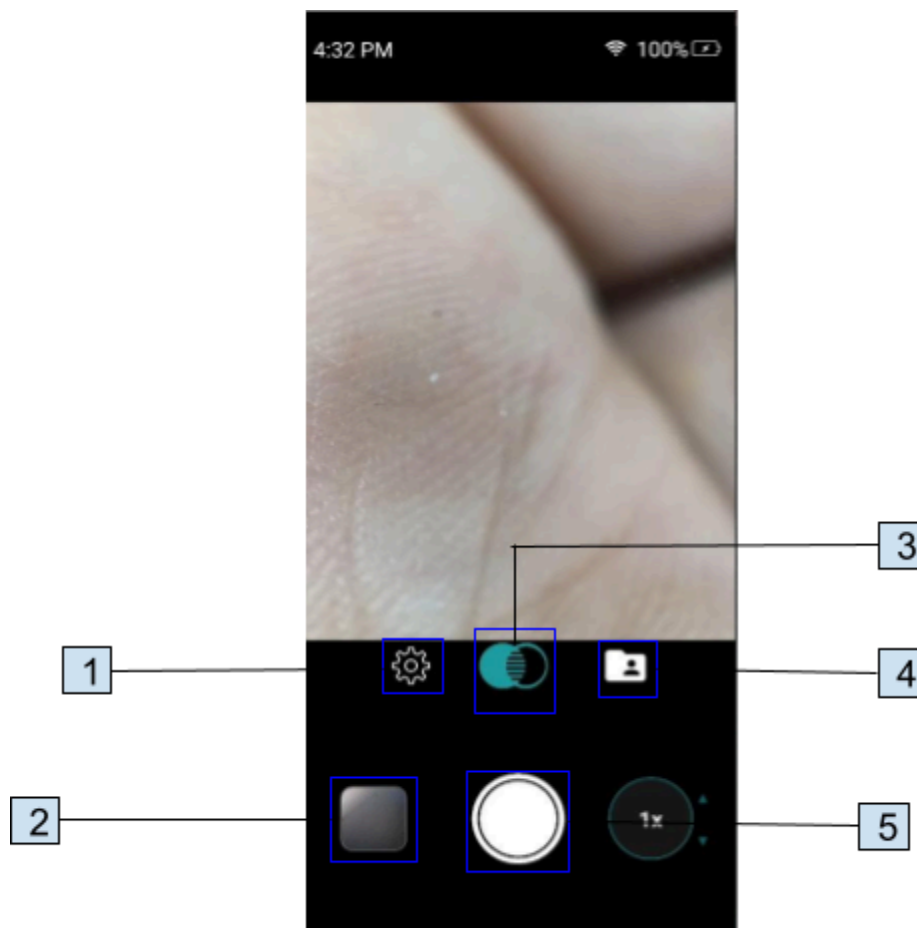
SESSION DATA PERMANENCE:

For patient privacy, all data captured during a Guest session is stored in volatile memory and is permanently erased upon session termination.

5.2. Camera Screen:

In the camera screen, the user can navigate to other screens, such as:

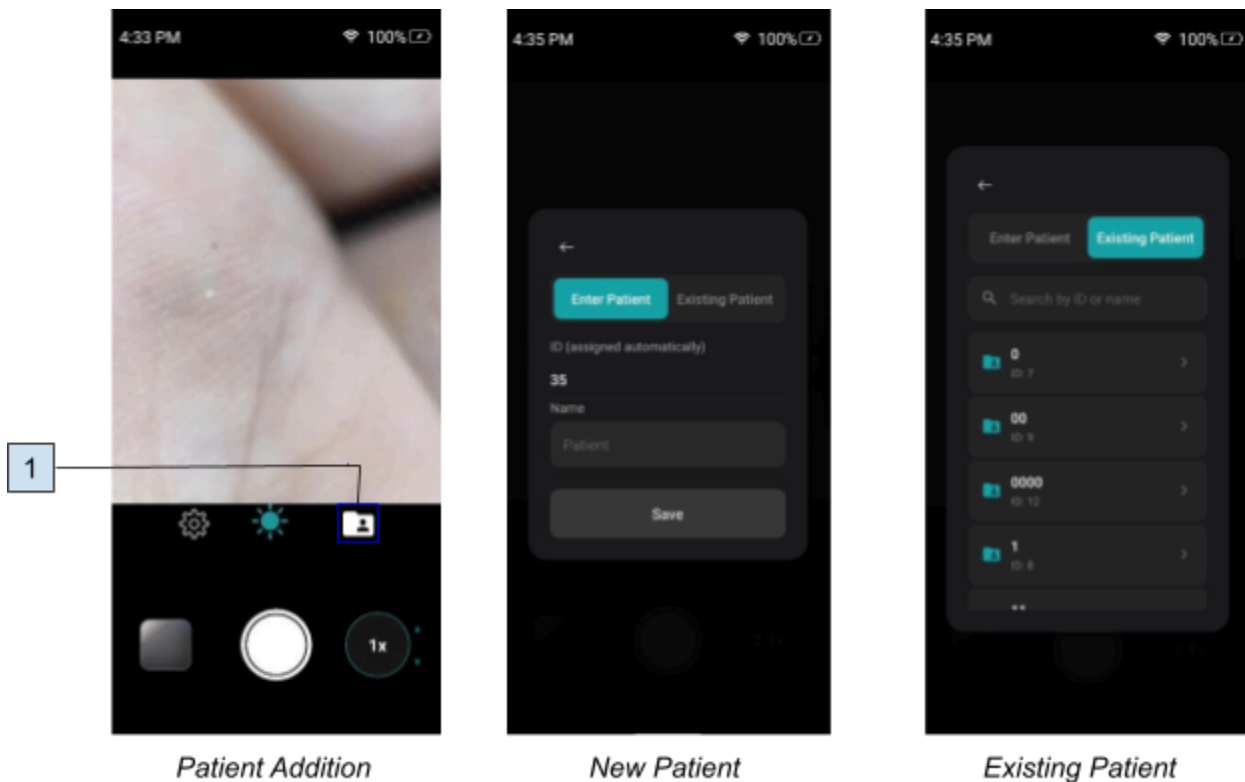
1. Image Capturing
2. Patient Addition
3. Gallery
4. Settings



1. Settings
2. Gallery
3. Polarization / Non-Polarization / Off State
4. Patient Addition / Folder selection
5. Image Capture

5.2.1. Patient Addition:

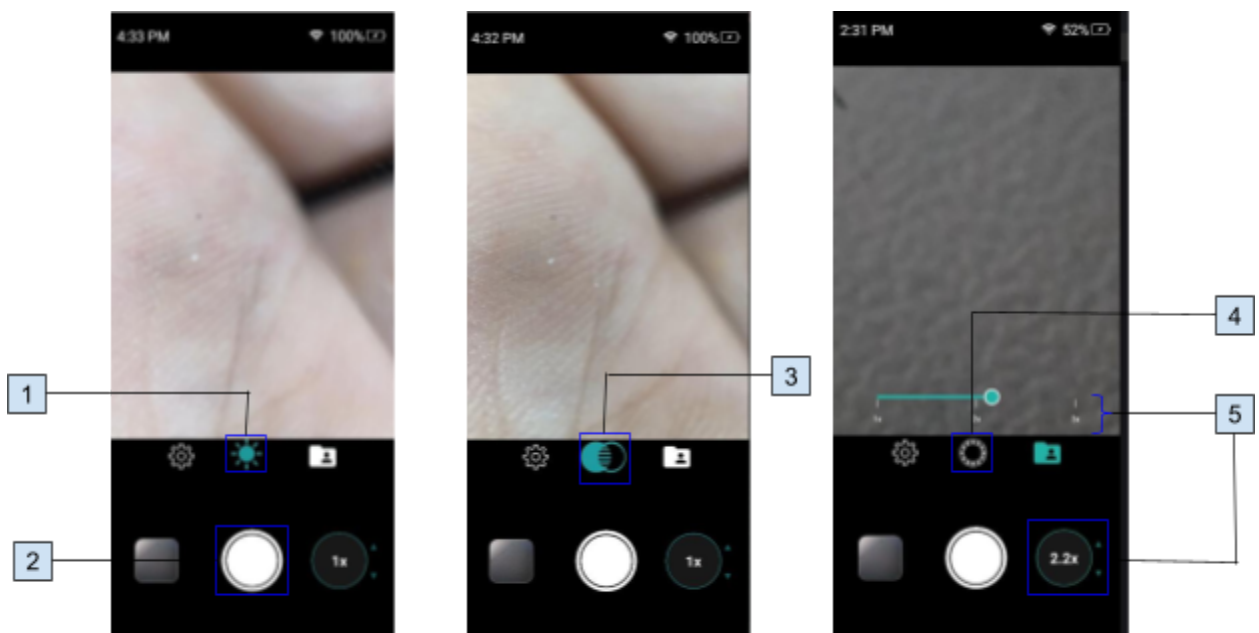
- When a clinician logs in, first he/she will add a new patient or select an existing one to capture an image.
- The captured image would be saved with the selected patient's name and in the folder created for that patient in the gallery to store captured images.



1.Patient Addition / Folder selection

5.2.2. Image Capturing:

- The image will be captured only if the user selects the patient from the list or enters a new patient as the captured images will be saved to that particular folder.
- Pressing the center button activates the Lens LED in polarized mode, accompanied by an on-screen indication.
- To toggle between polarized and non-polarized modes, press the lower button on the left side. The indication will update accordingly to reflect the selected mode.

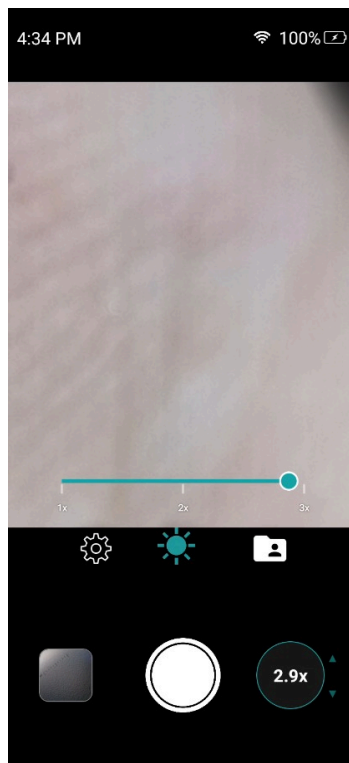


- 1.Non-Polarization
- 2.Capture Image
- 3.Polarization
- 4.Off State
- 5.Zoom Control

- Place the device gently on the skin.
- Adjust focus and magnification.

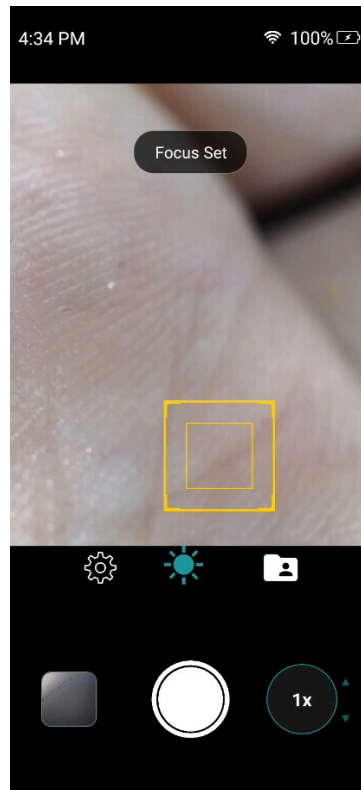
Zooming:

- Use the slider to fine control the zooming.
- Adjust magnification using on-screen gestures
- Use the slider to fine control the precision adjustment.



Focus:

- The lens has both auto focus and manual focus features.
- Tap the screen to focus manually.
- Capture images using a software interface.



5.2.3. Gallery:

- In the gallery, the images that are captured will be saved in the folder with the patient's name that clinician selects from the list or the new patient can be added.
- The images are uploaded to the cloud if the wifi is available; otherwise will be saved in the gallery but not uploaded to the cloud.

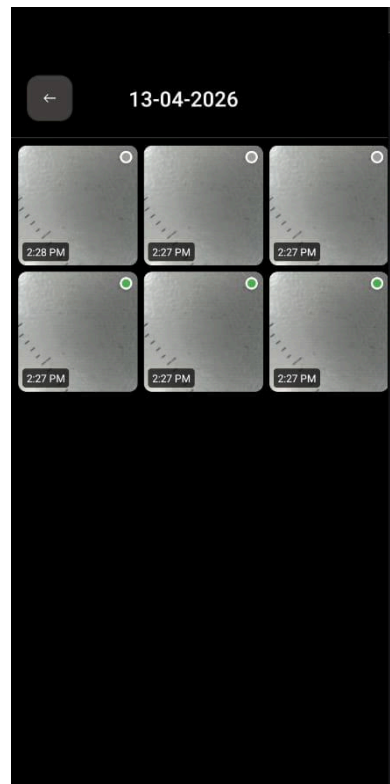
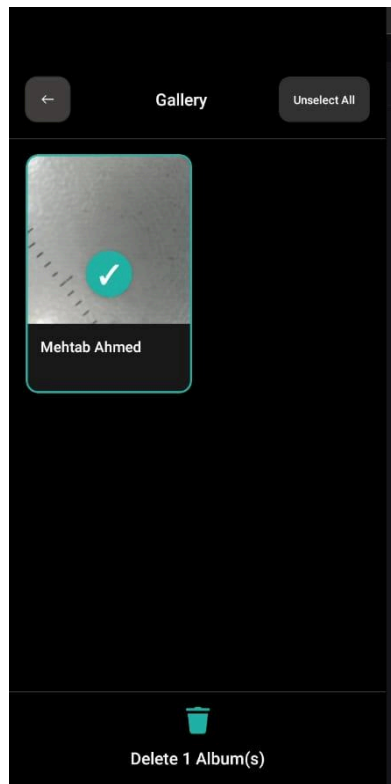


Image Uploading:

- Images that are captured are uploaded to the cloud if the Internet is available.
- If the Internet is not available, then the images are stored in the gallery, but the user will have to upload the image manually, either by selecting all at once or one at a time.
- The stored images are seen on the web app in the specific folder of the patient's name.
- The images that have a green indicator on the top-right corner are already uploaded to the cloud, and the remaining are not.

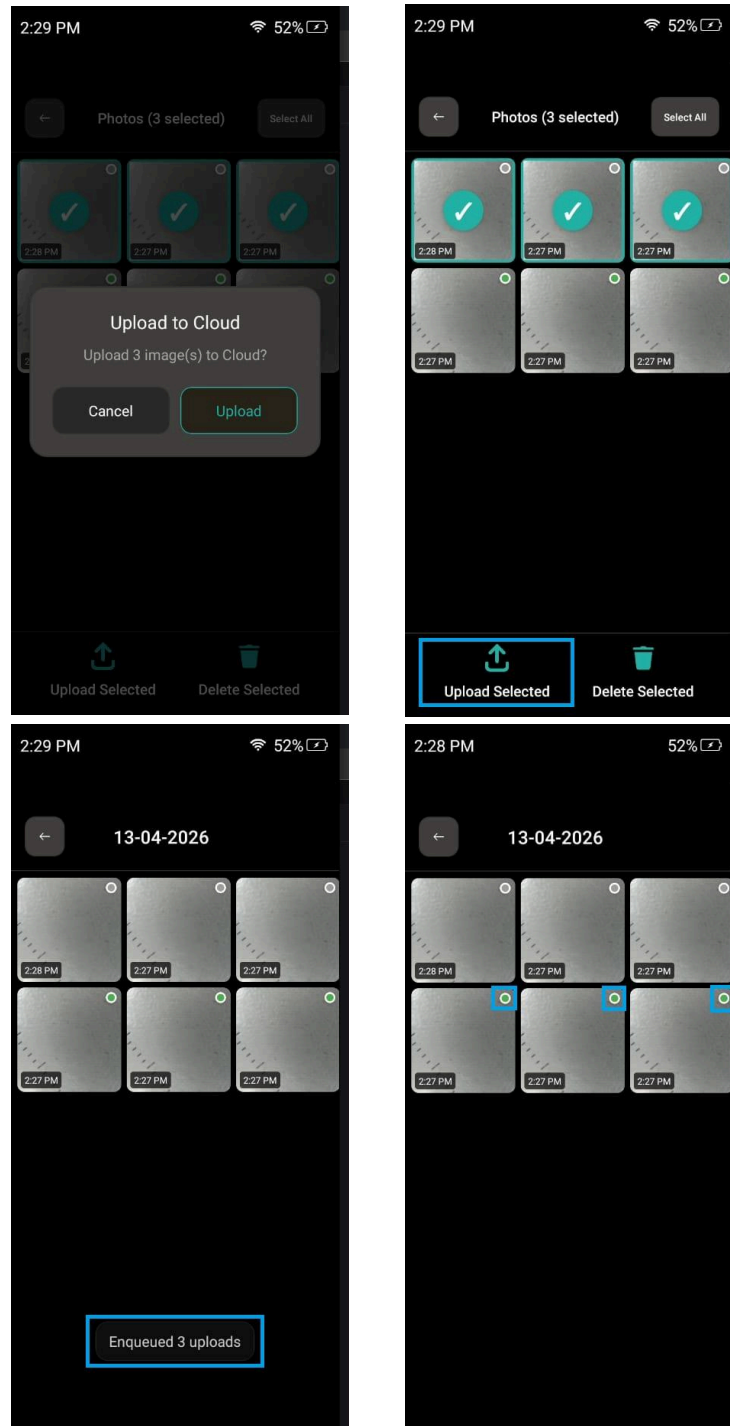


Image Deletion:

- Images that are captured are uploaded to the cloud if the Internet is available.
- If the Internet is not available, then the images are stored in the gallery, but the user will have to upload the image manually, either by selecting all at once or one at a time.
- The stored images are seen on the web app in the specific folder of the patient's name.
- The images that have a green indicator on the top-right corner are already uploaded to the cloud, and the remaining are not.

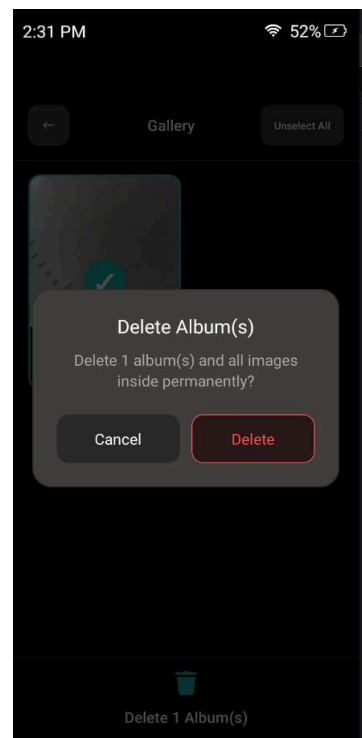
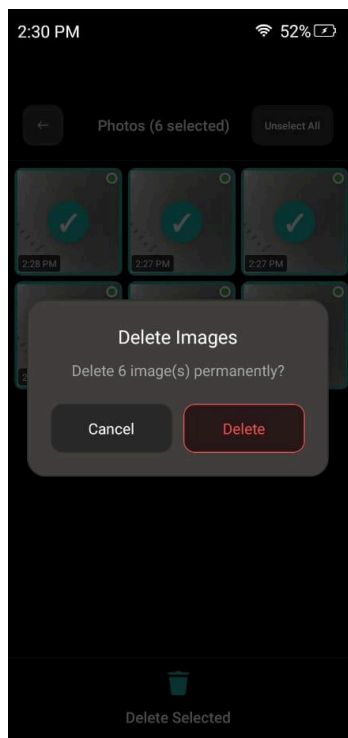
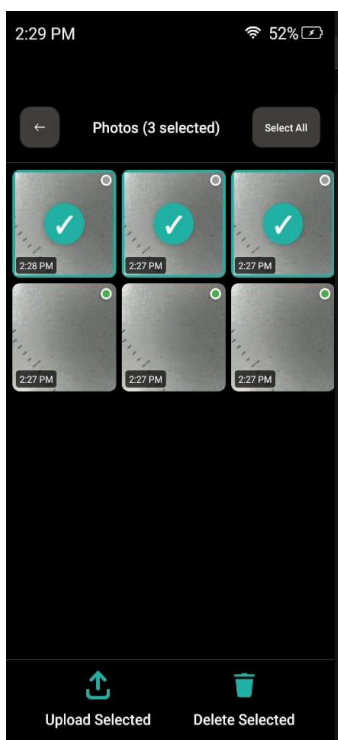


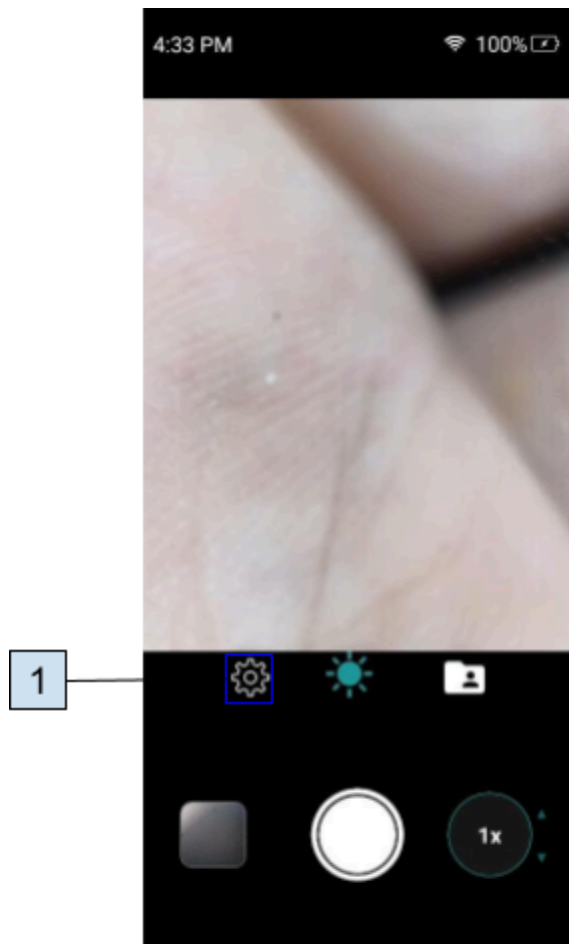
Image Viewing:

- View images on device screen
- Access stored images via patient/case manager
- Zoom and review captured data

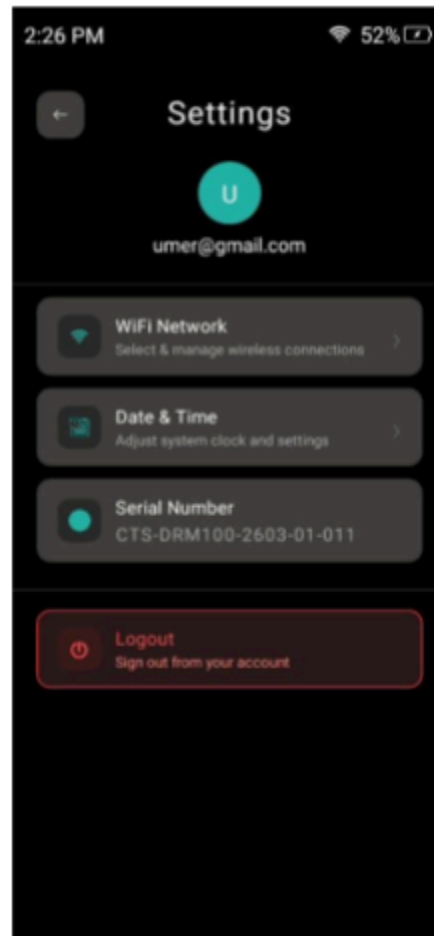


5.2.4. User settings:

- The user can set up their account, Wi-Fi, and date/time.
- The user can also log out via the logout button.



Camera screen

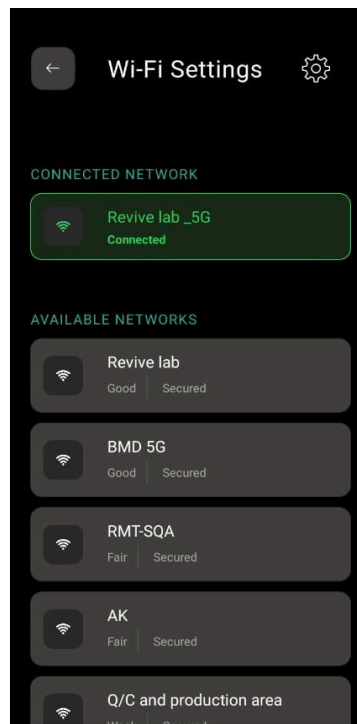


Settings

1. User Settings

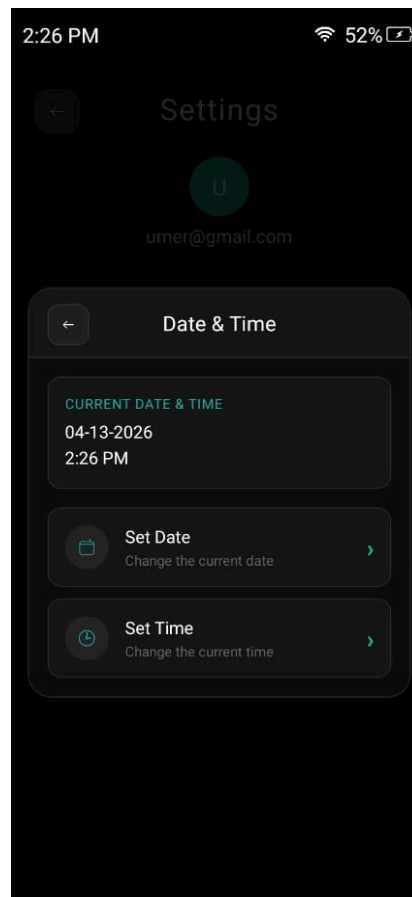
5.3 Wi-Fi Connectivity:

Automatically connects networks which are previously saved networks.



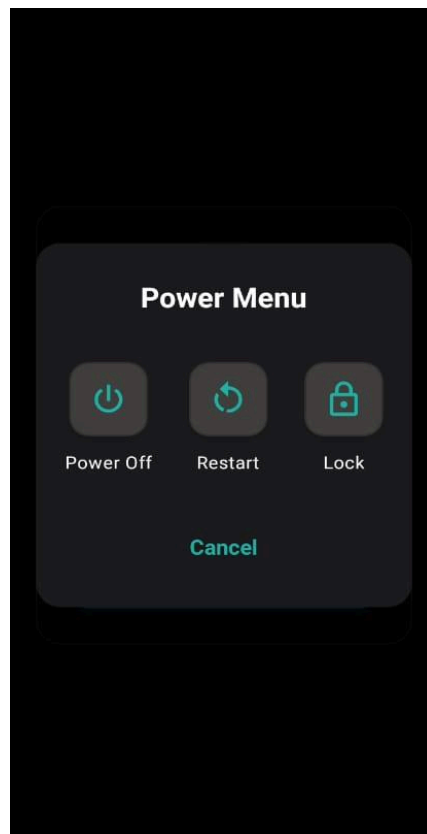
5.5 System Calibration (Date & Time):

- Accurate clinical documentation relies on precise chronometric data.
- The System Menu provides access to clock calibration.
- The CutiScope Dermatoscope maintains internal chronometric consistency across power Cycles.



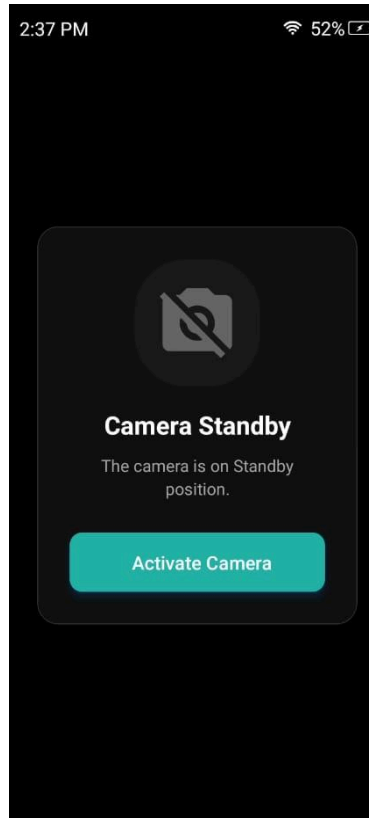
5.6 Power ON/OFF

- Long-press the power button to turn ON
- When the device is Powered ON, upon pressing the power button, the power menu appears so the user can:
 - Power OFF
 - Restart
 - Lock/Standby



5.7 Sleep Mode

The standby mode for the device is 2 minutes for no-touch input.



5.8 Battery Charging

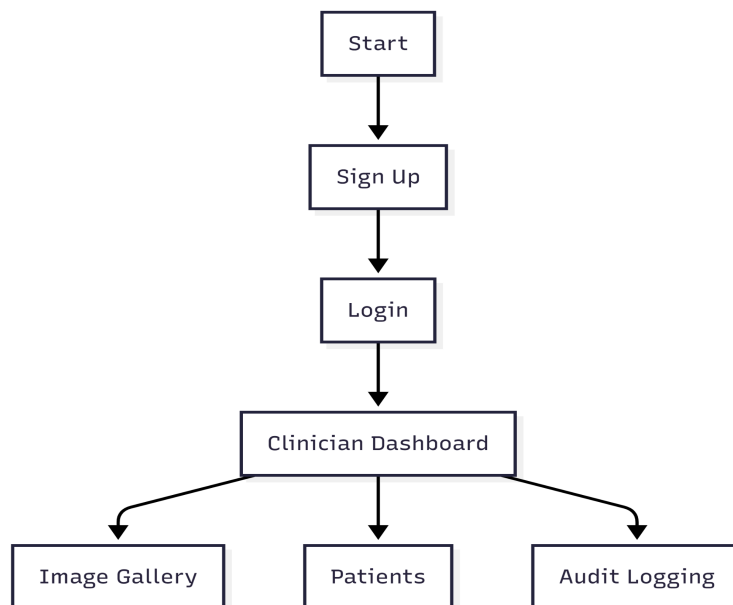
- Connect the USB cable to the device
- Charge via adapter
- The indicator shows the charging status.
- To ensure device longevity and data integrity, illumination is automatically moderated if the internal power level reaches a critical threshold (20%).

5.9 Web Dashboard:

The web dashboard is a secure, browser-based interface designed for clinicians to manage patient data and medical imagery. It connects to cloud storage and provides a central hub for viewing patients records, photos, and system activity.

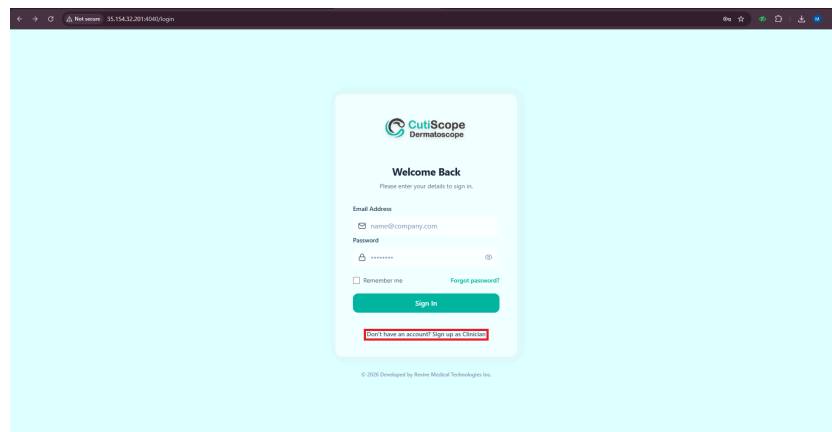
- View total patients, photos captured, and new patients weekly.
- Access organized folders named by patient ID or name.
- Find patients by name or ID, with details like creation date and photo count.
- See login history and user actions for security and auditing.
- Each clinician uses an email and password to access their account.

Operating Instructions:



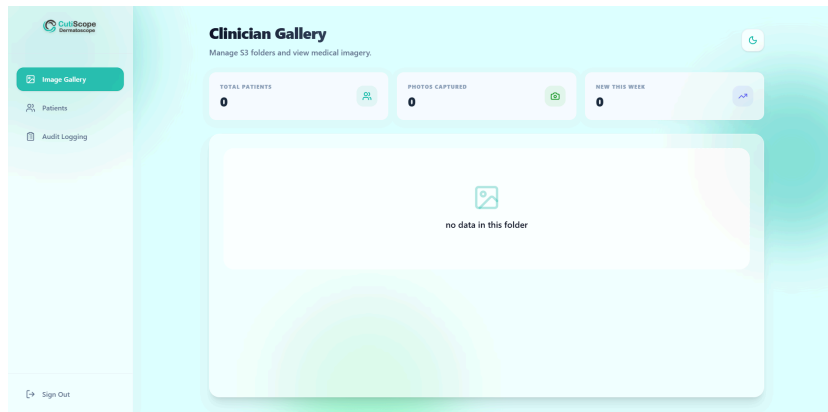
1. Sign Up /Log In:

- Visit the site: 35.154.32.201:4040/login



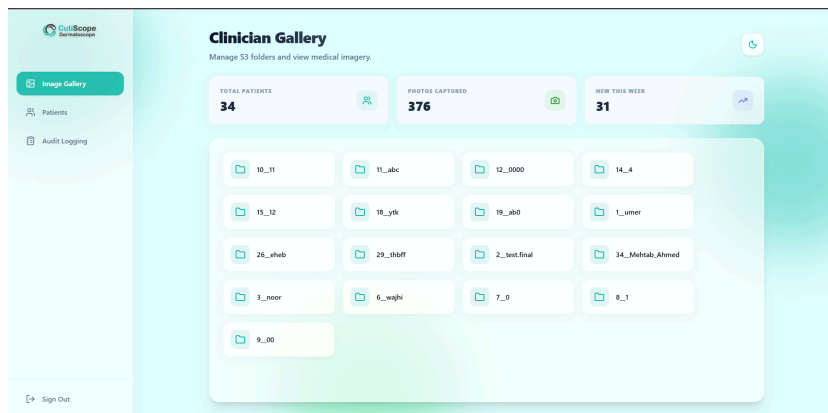
2. Dashboard:

After sign-up/login, the dashboard appears:



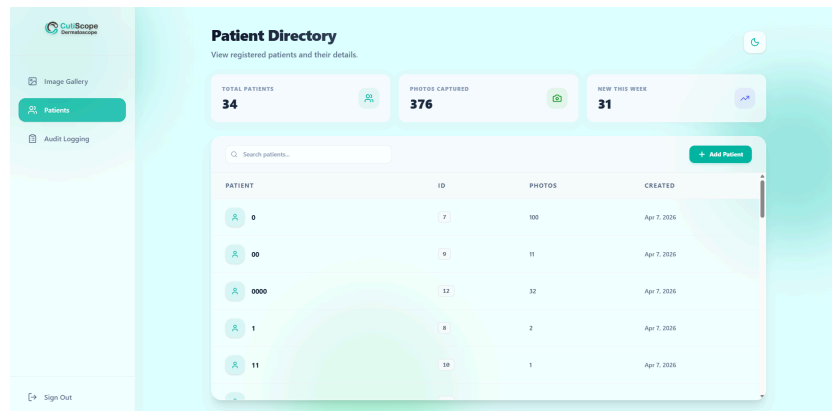
- **Image Gallery:**

In the image gallery, the Clinician can see all the images that he/she captured with the device.



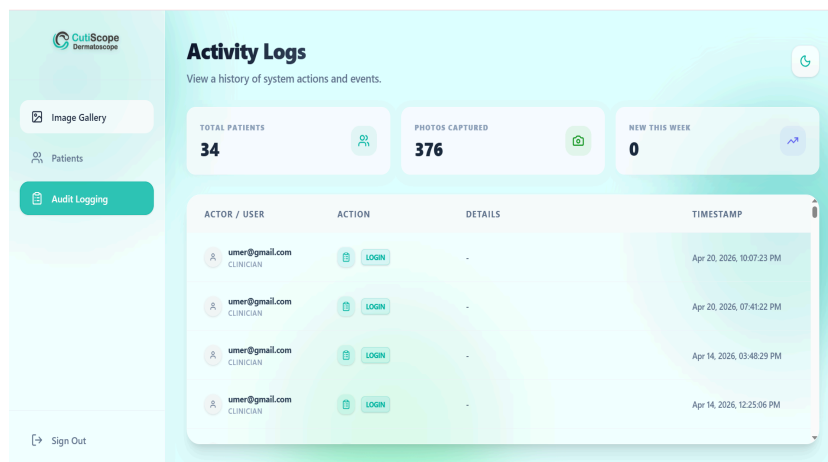
- **Patients:**

1. In the patients tab, the clinician can add the new patient.
2. The list of patients that are added or can also be seen.



- **Audit Logging:**

In the audit logging tab, the clinician can see the logs of the users added.



3.2. Handling & Use:

- Do not modify or disassemble the device.
- Use only accessories provided with the device.
- Do not immerse the device in liquid.
- Avoid dropping or mechanical shock.
- Ensure lens is not scratched or contaminated.
- Disinfect the device using only non-alcohol wipes. DO NOT USE IPA/ALCOHOLIC WIPES.
- Do not operate the device using wet hands
- Do not operate the device while it is charging
- Do not drop, shake, or expose the device to vibrations
- Do not use it for more than 30-40 minutes continuously to prevent the device from overheating.
- Do not unscrew the contact plate of the device.

3.3. Environmental and Operating Conditions

- If the room temperature rises rapidly, vapor may form on the lens; allow it to dissipate before use
- Do not operate or store the device in a dusty environment
- Do not operate or store the device outside the specified temperature limit of 35°C and the optimal humidity range.
- Do not operate or store the device near heating devices

3.4. Electrical Safety

- Do not attempt to open, modify, or repair the device, as this may expose internal electrical components.
- Use only the specified power adapter and the charging cable provided or recommended by the manufacturer.
- If the device shows signs of battery swelling, leakage, or overheating, discontinue use immediately.

- Keep the device away from strong electromagnetic sources such as high-frequency equipment.

5. Nomenclature

6. Device Features & Image Capture System

The CutiScope features a high-performance dual-mode polarization system for the surface.

- **Polarized Mode (Default):** Advanced cross-polarization optics eliminate surface glare, allowing for the visualization of deeper pigment networks, vascular patterns, and crystalline structures within the skin.
- **Non-Polarized Mode:** Utilized for the precise assessment of surface morphology, scale, and natural skin color.
- **Switching:** Use the push button to toggle between polarized mode and non-polarized mode.
- **Contact-based Dermoscopy:** This device supports only contact-based dermoscopy.

8. Clinical Workflow & Patient Management

8.1 Establishing a New Case

1. Access the Patient Manager via the folder icon.
2. **Clinical ID:** Input the unique patient identifier (Required for record-keeping).
3. **Reference Name:** Optional name for easier gallery navigation.
4. **Initialization:** The system automatically allocates a secure, encrypted directory for the new case.

8.2 Capture Information

Every image is automatically embedded with metadata, including:

- **Identifier:** Assigned Clinical ID.
- **Record Information:** Timestamp and the ID of the active clinician.

9. Data Storage & Synchronization

9.1 Structured Clinical Storage:

The CutiScope Dermatoscope utilizes an organized hierarchical filing system for all captures:

Case Folder → Year → Date.

Sync Status Indicators

- Green Marker: The data has been successfully verified and synchronized with the Clinical Cloud.
- Gray Marker: Synchronization is pending (ensure device connectivity).

9.2. Data Management:

- Enter the management mode by long-pressing any diagnostic record.
- Perform batch synchronization for pending records.
- Manage case directories with multi-step verification to prevent accidental data loss

11. CLEANING & MAINTENANCE

11.1. Cleaning

- Use a lint-free cloth
- Use non-alcohol wipes on external surfaces
- Do NOT apply liquid directly to the lens

11.2. Maintenance

- Inspect the device before use
- Do not open the device

12. Cleaning Protocol (Routine - Before/After Use)

Area	Frequency	Procedure
Contact Plate	Before & After Use	• Wipe with lint-free cloth using non-alcohol wipes on external surfaces; allow to air dry

Lens Surface	Before & After Use	Gently clean with soft optical cloth; avoid excessive pressure
Device Outer Body	Daily / As Needed	Clean with slightly damp cloth and approved disinfectant
Buttons	Daily	Wipe gently to remove dust and contaminants
Charging Port	Weekly / As Needed	Inspect and clean using dry air or soft brush (no liquid)
Full Device Check	Daily	Ensure no residue, moisture, or visible contamination remains

13. Troubleshooting Guide\

Problem	Possible Cause	Check / Corrective Action
Device not turning ON	Battery drained	Recharge device using USB-C charger
	Faulty cable	Replace with approved cable
	Internal fault	Contact manufacture
Poor image quality	Dirty lens/contact plate	Clean using proper cleaning procedure
	Improper positioning	Adjust focus and placement
	Wrong mode selection	Switch illumination mode
	Loose connection	Check cable and port

Device not charging	Damaged port/cable	• Replace or service
Overheating	Prolonged use	• Turn OFF and allow cooling
	High ambient temperature	• Use in recommended environment
Software not responding	Software glitch	• Restart application
Image not saving	Storage full	• Free up storage space
WIFI Not Connecting	Weak signal	• Restart the device • Move closer to the router • Check if networks are detected
Device Goes to Sleep Quickly		

Problem Observed	Possible Reason	Solution
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Device Not Powering ON	• Battery drained • Faulty charging cable or adapter • Power button failure	1. Charge for at least 30 minutes 2. Try another USB-C cable/adapter 3. Check charging indicator behavior
Battery Not Charging	• Damaged USB port • Battery degradation • Charging IC issue	1. Inspect the USB-C port for dust/damage 2. Verify adapter output (5V) 3. Check if the device heats up during charging 4. Reconnect the FPC connector if disconnected
No Image / Black Screen	• Camera module not initialized • Software freeze • Display issue	1. Restart the device 2. Check if UI is visible, but no image 3. Tap the screen to trigger focus 4. Try switching modes (polarized/non-polarized)

Wi-Fi Connecting	Not	• Weak signal • Antenna issue	1. Restart the device 2. Move closer to the router 3. Check if networks are detected at all 4. Check the antenna connection
Device Goes to Sleep Quickly		Normal Behavior: • Auto-standby after 120 seconds If abnormal: • Motion sensor not working properly	Press the Power Button If not → sensor or firmware issue
Lens LED Not Working		• Mode switching issue • Low battery protection (below 20%)	1. Switch polarization modes 2. Charge the device fully
Blurry Image / Focus Issue		• Dirty lens • Autofocus failure	1. Clean lens (lint-free cloth) 2. Use manual focus 3. Check for condensation

Display & Touch	• Black but ON → backlight failure • No touch → digitizer issue	1. Backlight voltage 2. Check the Display connector
System Standby	• The device is in power save mode.	• Press the power button to use the device.
Illumination Power Low	• Power level is <20%.	• Connect the Cutiscope Dermatoscope to a verified power source.
Pending Synchronization	• 	• Check the Wi-Fi Marker in the System Menu.
Initialization Refresh	• If the camera requires calibration, tap the "Wake Up" prompt.	

